

A Counterexample Without Quasi-Excellence

We record some more examples. See also [1, Examples 3.8–3.10].

Proposition. *The quasi-excellence hypothesis in the strong-generation theorem cannot be omitted, even for one-dimensional noetherian affine schemes.*

Proof. Let R be Hochster’s one-dimensional noetherian domain which is not J -0 [2, Example 1], and set $X = \operatorname{Spec} R$. Then R is neither J -1 nor quasi-excellent. If $D_{\operatorname{coh}}^b(X)$ had a classical generator, its image would classically generate $D_{\operatorname{sg}}(R)$, so Iyengar–Takahashi [3, Lemma 2.9] would imply that R is J -1, a contradiction. Thus $D_{\operatorname{coh}}^b(X)$ has no classical generator, hence no strong generator. $\square$

References

- [1] S. Dey and P. Lank, *Closedness of the singular locus and generation for derived categories*, arXiv:2403.19564v2 [math.AG], 2025.
- [2] M. Hochster, *Non-openness of loci in Noetherian rings*, Duke Math. J. **40** (1973), 215–219.
- [3] S. B. Iyengar and R. Takahashi, *Openness of the regular locus and generators for module categories*, Acta Math. Vietnam. **44** (2019), no. 1, 207–212.